ENGINEERING CHANGE NOTE (ECN)

PROJECT: Project Asynch
ORIGINATION: Professor Kid Kidd
PURPOSE OF CHANGE: Improving Reliability
DOCUMENT NUMBER: ECN 1.0
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EXECUTIVE SUMMARY: The system now includes an improved task management system. In essence, the program now includes a defined list of tasks and loops through them. Each task contains an interval that dictates when to run it, and when it was last run. All High Level Design requirements and Testing Requirements remain consistent.

HIGH-LEVEL DESCRIPTION: The system incorporates a round-robin cyclic executive task scheduler that utilizes a function pointer to go from task to task.

REASON FOR CHANGE: A cyclical round-robin system reduces labor and design costs (i.e a new task can be added easily)

Changed Documents:

- Design 2 [D2] as a section has been crossed out; replaced by D2*
- D1.1: Make Ports a ~~constant integer variable~~ #define'd macro
- D1.2: Define helper function ~~blink()~~ task_blink()

Added Documents:

- ECN 1.0 has been attached
- projCyclicTasking.INO has been attached
- Included Design 1.6 to 1.8 [D1.6, D 1.8]
- Included Design 4.1.1 to 4.1.4 [D4.1.1, D4.1.4]
- Included Design 5.1 to 5.3 [D5.1, D5.3]
- Included Design 6.1 [D6.1]

- Updated testing video

Removed Documents:

- asynchProj1.INO has been removed

Replaced Documents:

- Project 1: Asynchronous System has been replaced with Project 1.1: Cyclical System

Any Additional Design Considerations: The system now incorporates macros that utilize #define. This helps to reduce the data and memory consumption in the program. Once again all Testing and High Level Design requirements are consistent with Project Asynch.